

Literature review of mobile phone applications for forest inventory

Authors : Celine Mercier and Robin Hartley (Scion)

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Introduction

About this report

This report reviews mobile phone applications performing forest inventory tasks.

For each application (app), detailed information is reported, such as:

- App purpose (e.g. inventory, standing tree measurements, log measurements, pruning quality control, carbon/biomass estimation)
- If applicable, measurement types (e.g. height, diameter at breast height (DBH), stem volume)
- Species and forest type the app was developed for, or tested on
- Ease of use (e.g. is the app well documented, can it be easily used by anyone)
- When possible, assess app accuracy (through literature or app reviews)
- Platform and requirements (Android/iOS, phone with lidar)
- App popularity (e.g. download count)
- App age and maintenance (e.g. time of release and last update, is it possible to communicate with the developers)
- Country where the app was developed, and insight into the company (e.g. developed by software company, forestry company, university)
- App licence status and cost

Method for app review

Apps and related information were found by searching the Android and Apple application stores, the general web with classic search engines, as well as internet forums and social media such as Reddit (www.reddit.com) where forestry stakeholders discuss their choices and use of different apps.

To gain insights into the usability of the most prominent apps, the authors briefly tested them in a mature stand of coast redwood (*Sequoia sempervirens*), using an iPhone 13 Pro Max, or a Google Pixel 5. This stand is located in the Redwood Memorial Grove, opposite Scion's research arboretum. The stand is ~124 years old, and the average tree height is 67 m, with many trees having a diameter in excess of a metre.

About the different types of apps and the structure of this report

Two main types of apps exist for forest inventory. The first type uses phone technologies such as lidar, Augmented Reality (AR), or computer vision methods that have emerged in recent years to perform tree or log measurements. The second type only enables manual entry of information, sometimes with the ability to also register

the coordinates or picture of a tree through the phone's GPS or camera. This report is divided in three parts, exploring the first, more complex category in greater depth:

1. Applications that use phone or computer vision technologies for tree or forest measurements (figure 1),
2. Applications that use phone or computer vision technologies for log pile measurements (figure 2),
3. Applications that facilitate manual entry for forest inventory (figure 3).

Within each part of the report, apps are sorted alphabetically.



Figure 1. Example of app that uses phone or computer vision technologies for forestry measurements. Here, Arboreal Forest uses lidar to measure diameters, and AR to display plot border (in blue) and measurements.

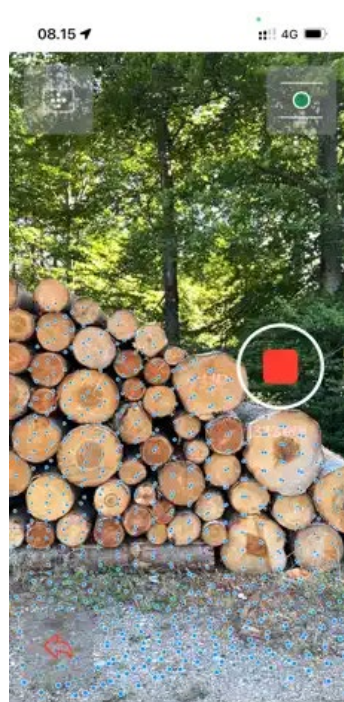


Figure 2. Example of an app that uses phone or computer vision technologies for log pile measurements. Here, the app LogStackLIDAR uses lidar to measure individual logs as well as log piles.



Figure 3. Example of an app that facilitates manual entry for forest inventory. Here SilvAssist Mobile is used to enter cruising data.

General notes

All apps in this report are available in English and in New Zealand. The Apple store does not offer information on the download count (and therefore popularity) of its apps.

Most of the apps in this report use the global navigation satellite system (GNSS) in addition to the other technologies listed, to save the coordinates of plots and/or trees.

Some apps performing 'inventory-adjacent' tasks were not included in this report, for example, the BasalArea app, which computes basal area using the phone camera (https://play.google.com/store/apps/details?id=com.forest.basalarea&hl=en_NZ).

Other 'apps' not included in this report, are in reality web apps (i.e. web pages) that can be accessed from mobile phones ; e.g. <https://planitgeo.com/treeplotter-inventory>, an app designed for the manual entry of urban tree information by arborists. Finally, some apps exist for general GIS inventory purposes, such as ArcGIS Field Maps and Survey123, both developed by ESRI. These latter two apps are frequently used by the forestry industry for various inventory tasks.

1. Applications that use phone or computer vision technologies for tree or forest measurements

Table 1. Overview of the applications that use phone or computer vision technologies for tree or forest measurements.

Name	Platform	Developer	Technologies used	Country	Year released	Year last updated
Arboreal Forest	Apple	Arboreal	AR/Camera, lidar if available	Sweden	2019	2025
Arboreal Tree Height	Android, Apple	Arboreal	AR/Camera, lidar if Apple	Sweden	2018	2025 (Android), 2024 (Apple)
Cruise Angle iCMT	Apple	Corvallis Microtechnology, Inc.	AR/Camera	US	2013	2024
ForestScanner	Apple	MAPRY Co., Ltd.	AR/Camera, lidar	Japan	2021	2022
iForester	Apple	Purdue University Institute for Digital Forestry	AR/Camera, lidar	US	2023	2025
KATAM Forest: Decision Support	Android	KATAM Technologies AB	Camera	Sweden	2019	2024
MOTI	Apple	Bern University of Applied Sciences	AR/Camera	Switzerland	2015	2016
Trees	Android	Forest Monitoring Tools	AR/Camera	Japan	2019	2024
Trestima	Android, Apple	Trestima Oy	Camera	Finland	2014 (Android), 2016 (Apple)	2023 (Android), 2025 (Apple)

1.1. Arboreal Forest

- **Website:**
 - <https://arboreal.se/en/arboreal-forest>
 - <https://apps.apple.com/nz/app/arboreal-forest/id1462673493>
- **Platform:** Apple
- **Recommended devices:** iPhone 12 Pro and later for the lidar sensor
- **Owner:** Arboreal AB
- **Technologies used by phone to perform measurements:** AR/Camera, lidar if available, Inertial Measurement Unit (IMU; a suite of sensors including a gyroscope and an accelerometer that is used by the phone to calculate its position in 3D space)
- **Country:** Sweden
- **Year:** 2019
- **Last updated:** 2025
- **Cloud storage:** Yes, within the EU
- **Works offline:** Yes
- **Pricing:** Free trial (5 plots), then \$17.49 NZD per week or \$54.99 NZD per month, purchased in-app
- **Measurements/predictions:** Diameter, height, basal area, Quadratic Mean Diameter, trees per ha, standing volume per ha, solid volume per ha, volume per tree, mean canopy height, CO₂ captured
- **Ease of use and documentation:** The official website is easy to navigate. Several tutorial videos are available. The app also displays instructions and indicates success or issues.
- **Comments:**

This is one of the three forest inventory apps developed by Arboreal, a private company based in Sweden.

This app uses AR and lidar technology to assist in the inventory and measurement of trees in a circular plot, along a line transect, or in an open plot. AR is used to position the plot and display its limits (figure 4). The user points at trees, and either AR or lidar is used to measure the DBH. Height is also measured, using AR and internal sensors.

Height measurement requires the user to be at least half of the tree height away from the tree, e.g. if the tree is 30 metres high, the user needs to be at least 15 metres away from it, and see the top, to perform the measurement accurately. The app, through AR, can show with a red tag which tree should be used to calculate Lorey's height. The app then infers a plot-level estimate of basal area, Quadratic Mean Diameter, trees per ha, standing volume per ha, solid volume per ha, volume per tree, mean canopy height, and CO₂ captured (see figure 5 for an example of a result table).



Figure 4. Screenshot of the Arboreal Forest app while taking measurements. The border of the plot is visible in blue. The computed diameter of a tree is displayed. Image taken from the Arboreal Youtube channel (https://www.youtube.com/watch?v=e6fkN_t0bQ).

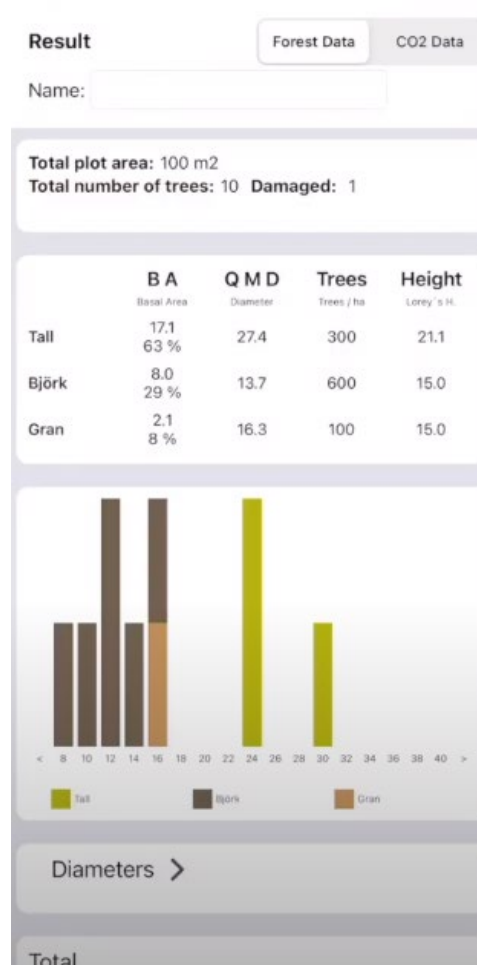


Figure 5. Screenshot of the Arboreal Forest app showing some of the plot measurement results. Image taken from the Arboreal Youtube channel (<https://www.youtube.com/watch?v=nQRxtD7Y4Dc>).

The region and tree species must be selected from a list, and the species can be changed for each tree measured. The list includes radiata pine (*Pinus radiata*) and douglas fir (*Pseudotsuga menziesii*), and the company offers to add any species not already in the list, as needed.

According to several scientific articles and reports, the app is accurate for DBH measurements ^(1, 2, 3, 4, 5, 6).

A study by Sandim et al. comparing three smartphone applications for forest inventory ⁽¹⁾ found that the app was overall accurate, and the most accurate when compared to the Katam and Trestima apps. It performed especially well in adult maritime pine (*Pinus pinaster* Aiton) forest “without under-cover regions and without slope”. Another study done in Costa Rica ⁽²⁾ found that the DBH measurements were overall accurate, except for thin stems (less than 5 cm). They also reported that the height estimation was challenging to perform and less accurate in sloped terrain, and in forests with heavy understory. This is not surprising, as the app was primarily developed for, and tested in, the relatively flat and low-density forests of Sweden.

Tests done by the authors with an iPhone 13 Pro Max in a coast redwood forest found that the app was overall easy to use, but the height measurement was challenging, as these trees were exceptionally tall (60 m+) with a closed canopy, making it difficult to see individual tree peaks. The app also failed to detect large stems and calculate their diameters (when larger than about 1.5 m), although trees of that size are not typically of interest for New Zealand foresters.

1.2. Arboreal - Tree (Apple) / Arboreal Tree Height (Android)

- **Website:**
 - <https://arboreal.se/en/tree-height>
 - https://play.google.com/store/apps/details?id=se.arboreal.height&hl=en_NZ
- **Platform:** Android, Apple
- **Recommended devices:** iPhone 12 Pro and later for the lidar sensor; any Android phone with AR capability (available in most phones released in recent years, https://developers.google.com/ar/devices#google_play)
- **Owner:** Arboreal AB
- **Technologies used by phone to perform measurements:** AR/Camera, lidar if available, IMU
- **Country:** Sweden
- **Year:** 2018
- **Last updated:** 2024 for Apple version, 2025 for Android version
- **Popularity:** 50k+ downloads on the Google Play store (Android)
- **Cloud storage:** Yes, within the EU

- **Works offline:** Yes
- **Pricing:** Free trial (5 measurements), then \$59.99 NZD for unlimited measurements in the Android version; on the Apple version, \$4.99 NZD per month, \$19.99 NZD per year, or \$49.99 NZD for unlimited measurements. All purchased in-app.
- **Measurements/predictions:** Tree height, crown height, crown width, tree inclination, diameter if Apple phone with lidar, tree coordinates.
- **Ease of use and documentation:** The official website is easy to navigate. Several tutorial videos are available. The app also displays instructions and indicates success or issues.
- **Comments:**

This is another of the three forest inventory apps developed by Arboreal, a private company based in Sweden.

This app uses AR, as well as the camera and internal sensors to perform measurements of individual tree heights, inclination and crown dimensions. If a lidar sensor is available, the Apple version also measures the tree's diameter. The user first approaches a tree stem, at a distance of about 1 to 2 meters; the app must detect the stem. The user then uses AR to tag the base, then walks away from the tree enough to see the top, ideally at a distance equivalent to the height, and tags the base again as well as the top and the crown base and sides, following on-screen instructions (figure 6). The tree height, inclination and crown dimensions are then displayed and saved (figure 7). Pace et al. (7) found the app measurements to have a strong correlation with measurements carried out using field instruments.

Tests done by the authors with an iPhone 13 Pro Max in a coast redwood forest found the same results as for the Arboreal Forest app previously described: the app was overall easy to use, but the height measurement was challenging as these trees were exceptionally tall (60 m+) with a closed canopy, making it difficult to see individual tree peaks. Walking 20 m away from a tree often means not seeing its base or top anymore, depending on the forest type. The app also failed to detect large stems and calculate their diameters (when larger than about 1.5 m), although trees of that size are not typically of interest for New Zealand foresters.



Figure 6. Screenshot of the Arboreal Tree Height app while taking measurements. The tree top has to be visible enough to mark it. Taken from the Arboreal Youtube channel (<https://www.youtube.com/watch?v=w9kbmwffue4>).



Figure 7. Image of the Arboreal Tree Height app displaying measurement results.

1.3. Cruise Angle iCMT

- **Website:**
 - <http://www.cmtinc.com/>
 - <https://apps.apple.com/nz/app/cruise-angle-icmt/id620252916>
- **Platform:** Apple
- **Recommended devices:** iOS 9.0 or later
- **Owner:** Corvallis Microtechnology, Inc.
- **Technologies used by phone to perform measurements:** AR/Camera
- **Country:** US
- **Year:** 2013
- **Last updated:** 2024

- **Pricing:** \$19.99 NZD
- **Comments:**

This app is mainly designed for manual entry while cruising, but includes an angle gauge through AR (figure 8):

When using the point sampling method for timber inventory, the forester first determines a suitable Basal Area Factor (BAF) for the timber stand. Then the corresponding angle gauge is used to select the sample trees. At each sampling point, the forester scans the trees through the opening of the angle gauge. Any tree that completely fills the opening is deemed a sample tree. From the tally of sample trees, one can obtain the basal area per acre.



Figure 8. Screenshot of the Cruise Angle iCMT app while taking measurements. The gauge is visible in blue. Taken from the app page on the Apple store.

1.4. ForestScanner

- **Website:**

- <https://apps.apple.com/us/app/forestscanner/id1599055402?l>

- <https://doi.org/10.1111/2041-210X.13900>

- **Platform:** Apple

- **Recommended devices:** iPhone 12 Pro and later for the lidar sensor

- **Owner:** MAPRY Co., Ltd.

- **Technologies used by phone to perform measurements:** Lidar, AR/Camera

- **Country:** Japan

- **Year:** 2021

- **Last updated:** 2022

- **Pricing:** Free

- **Works offline:** Yes, once an account is created and activated.

- **Measurements/predictions:** 3D point clouds, stem diameters and spatial coordinates based on real-time instance segmentation and circle fitting

- **Ease of use and documentation:** The developers published a scientific paper describing the app ⁽⁸⁾. The app is intuitive and easy to use. However, obtaining a high-quality point cloud for accurate stem detection and diameter estimation seems to require some time testing optimal distances, trajectories, orientations, etc.

- **Comments:**

The user scans the environment with the app, and ForestScanner estimates the stem diameters and spatial coordinates based on real-time instance segmentation and circle fitting. Tree diameters can be displayed and comments entered for each tree while scanning. The user can then visualise and share the results. Point clouds can be exported in PLY format, and tree information (diameters and coordinates) can be exported in CSV format.

Several studies ^(2, 6, 8, 9) found that the DBH estimations were accurate. Howie et al. found that accurate measurements required a relatively clear understory ⁽⁶⁾.

Tests done by the authors with an iPhone 13 Pro Max in a coast redwood forest found that the app was overall easy to use, and the display of the scanned scene was impressive, both while scanning (figure 9) and once the scan was finished (figure 10). However, at some point during the scan, the app seemed to lose track of the movements and objects, and created a lot of artefacts and duplicated objects in the

scanned scene from that point on (see reconstructed scene figure 10). This could, however, be due to a user error e.g. pointing the camera at the ground, which hinders AR tracking.

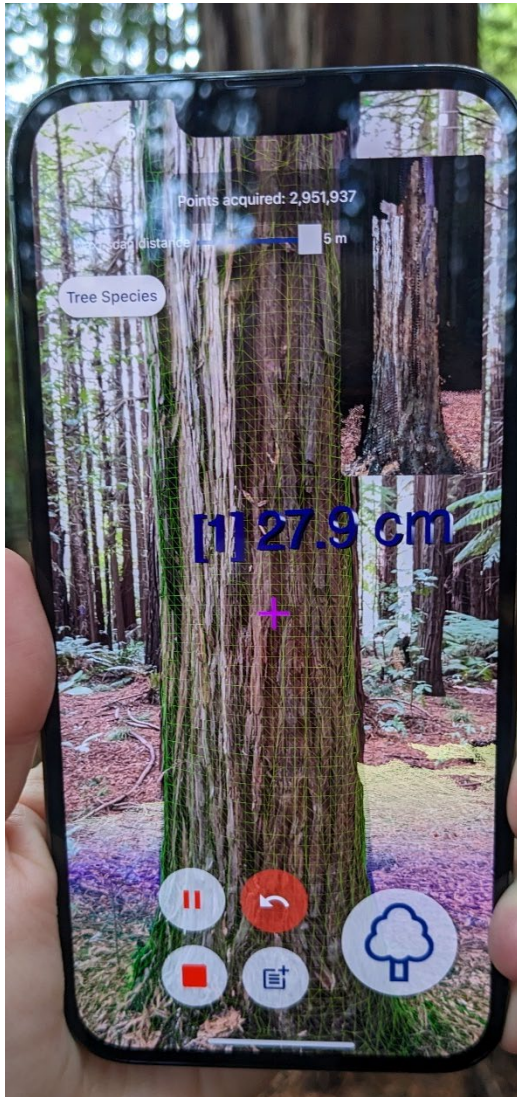


Figure 9. Image of the ForestScanner app while taking measurements. A model of the scene is visible in the top right corner. The tree diameter computed on-the-fly is also displayed.

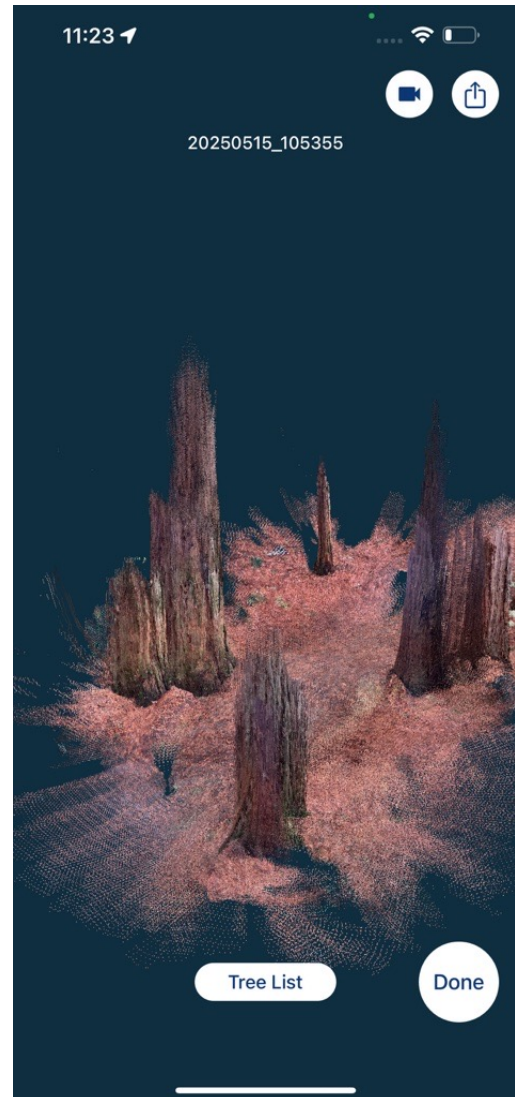


Figure 10. Screenshot of the ForestScanner app displaying the reconstructed scene. Artefacts are visible (double trees).

1.5. iForester

- **Website:**

- <https://digitalforestry.org/forester/>
- <https://apps.apple.com/nz/app/forester/id6464708769>

- **Platform:** Apple
- **Recommended devices:** iPhone 12 Pro and later for the lidar sensor
- **Owner:** Purdue University Institute for Digital Forestry
- **Technologies used by phone to perform measurements:** Lidar, AR/Camera
- **Country:** US
- **Year:** 2023
- **Last updated:** 2025
- **Pricing:** Free
- **Cloud storage:** Yes
- **Works offline:** Yes
- **Measurements/predictions:** Stem DBH and diameter at any height from individual trees
- **Ease of use and documentation:** The app is very easy to use. The website is also a good source of information.
- **Comments:**

This app uses AR and lidar, if available, to calculate the DBH, or diameter at any height, of individual trees. It can display either a normal camera view, or a "3D" view showing the depth perception of the lidar, facilitating the more challenging measurements. It uses the Segment Anything Model (SAM) for tree detection from the image, and the lidar for the depth and distance perception used for diameter computation (<https://htirc.org/forester-ai-assisted-smartphone-app-for-automated-tree-inventory/>). The developers are now working on adding height estimation. They then plan to work on tree grading algorithms for evaluating log values using DBH and height information, as well as species classification based on tree surface vein patterns and size data.

Tests done by the authors with an iPhone 13 Pro Max in a coast redwood forest found that the app was very easy to use and seemed able to detect and measure the diameters of even the larger trees (about 1.8 m), unlike the Arboreal apps.



Figure 11. Image of the iForester app successfully measuring the DBH of a large redwood tree.

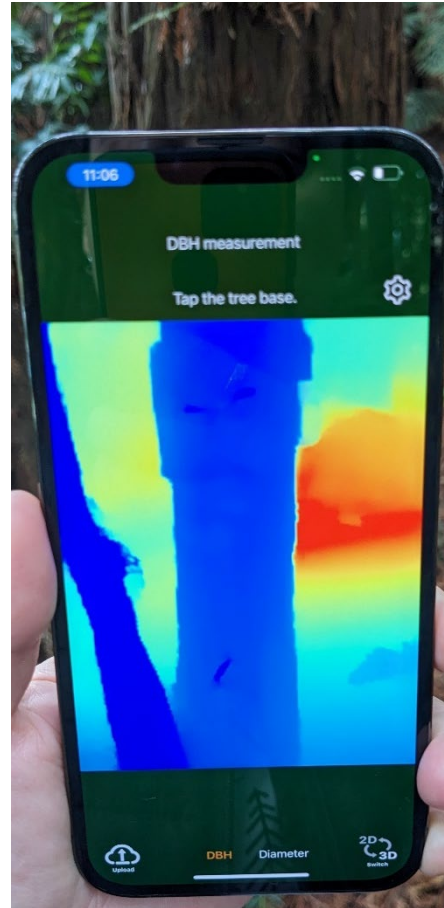


Figure 12. Image of the iForester app while taking measurements in "3D view" mode, showing the depth perception of the lidar to facilitate accurate measurements.

1.6. KATAM Forest: Decision Support

- **Website:**
 - <https://www.katam.se/>
 - https://play.google.com/store/apps/details?id=katam.com.datarecorder&hl=en_NZ
- **Platform:** Android
- **Recommended devices:** Most recent Android phones seem compatible (<https://www.katam.se/tutorials/interoperability>)
- **Owner:** KATAM Technologies AB
- **Technologies used by phone to perform measurements:** Camera
- **Country:** Sweden
- **Year:** 2019
- **Last updated:** 2024

- **Pricing:** 30 day trial, 30 euros / month, 300 euros / year
- **Cloud storage:** Yes
- **Works offline:** Yes
- **Measurements/predictions:** Average diameter, average volume, basal area, stem density, diameter distribution; tree position, diameter, height and volume
- **Ease of use and documentation:** The official website is easy to navigate. Several tutorial videos are available. The company offers training with experts.
- **Comments:**

The app works by having the user walk or drive around while taking a video of the forest. The app detects all stems in the video and computes diameters (figure 13, 14, 15, 16). It then infers the volume, basal area, stem density. The average height can be entered manually.

According to two studies ^(1, 10), the accuracy of the app depends on the type of forest. The app produces more accurate results in forests with straight stems of average size with sparse understory.

Tests done by the authors with a Google Pixel 5 in a mature coast redwood forest found similar results, where thinner and larger stems were not correctly detected by the app (figure 13), although trees of diameters greater than a metre, as found in this stand, are not typically of interest for New Zealand foresters.

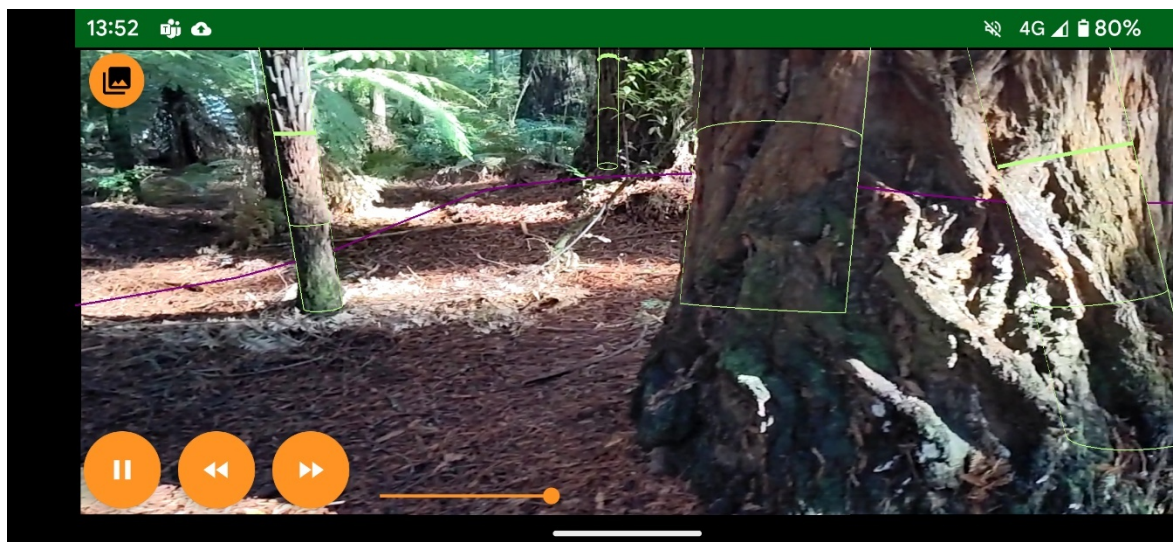


Figure 13. Screenshot of the Katam app displaying the detected tree stems after video processing. Erroneous detections are visible in large redwood stems. The correct detection of a smaller stem is also visible.



Figure 14. Screenshot of the Katam app displaying the shape of the filmed area (in purple), as well as the user's track (in red), and the detected stems (in green) with sizes corresponding to their computed diameters.

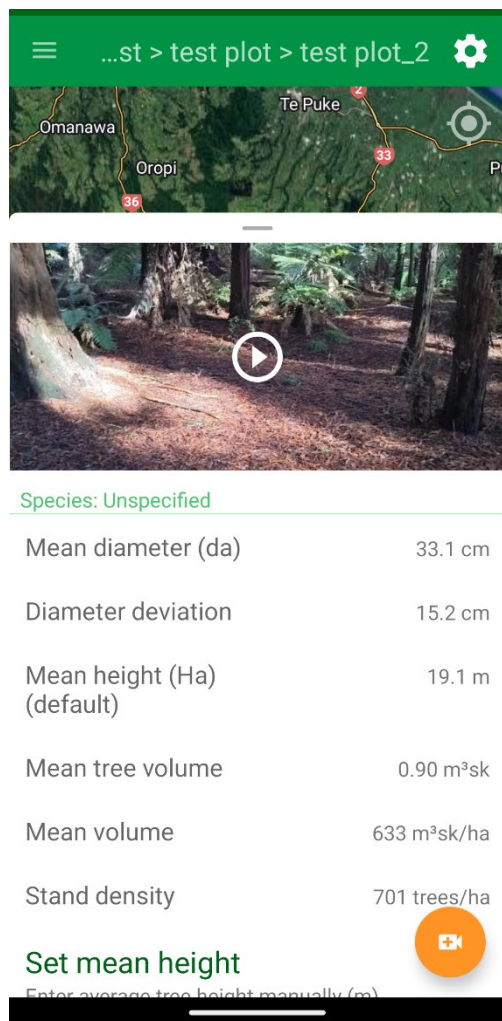


Figure 15. A screenshot of the Katam app displaying the results computed from the video.

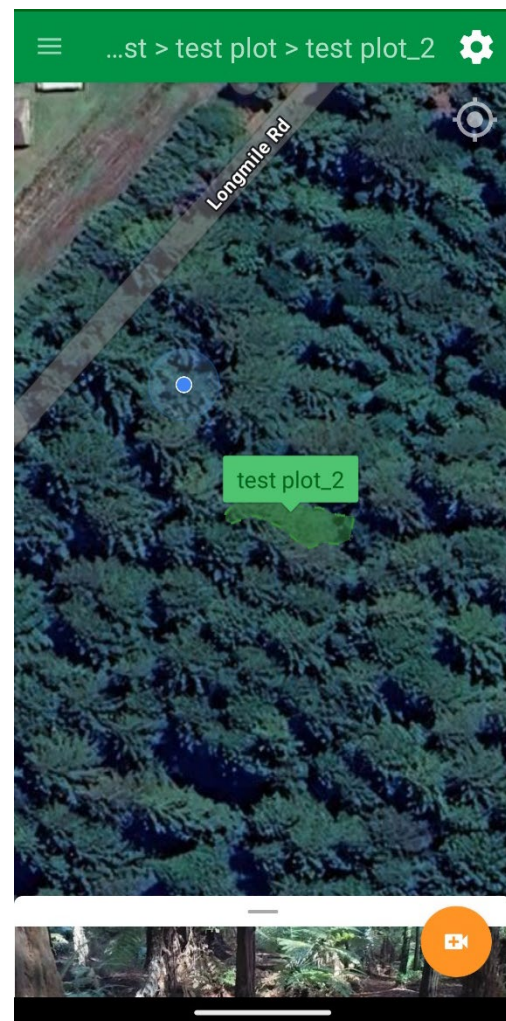


Figure 16. A screenshot of the Katam app displaying the position and shape of the plot where trees were filmed and detected.

1.7. MOTI mobile timber cruise

- **Website:**

- <https://www.moti.ch/drupal/>
- <https://apps.apple.com/us/app/moti-mobile-timber-cruise/id1039331308>

- **Platform:** Apple
- **Recommended devices:** iOS 8.0 or later
- **Owner:** Bern University of Applied Sciences
- **Technologies used by phone to perform measurements:** AR/Camera
- **Country:** Switzerland
- **Year:** 2015 on the Apple Store
- **Last updated:** 2016 on the Apple Store
- **Pricing:** Free
- **Cloud storage:** No
- **Measurements/predictions:** Basal area per ha, stems per Ha, tree height
- **Ease of use and documentation:** The available documentation on the website is not very helpful, and the app is not intuitive and easy to use.
- **Comments:**

Similar to the Cruise Angle iCMT app, MOTI is mainly designed for manual entry while cruising, but acts as a relascope and includes different gauges through AR to facilitate selecting and measuring trees (figure 17).

A study by Ficko ⁽¹¹⁾ reported results similar to those of Trestima, also described in this report.

The app seems deprecated and has not been updated for 9 years. It also used to be available on the Google Play Store (Android), but not anymore.

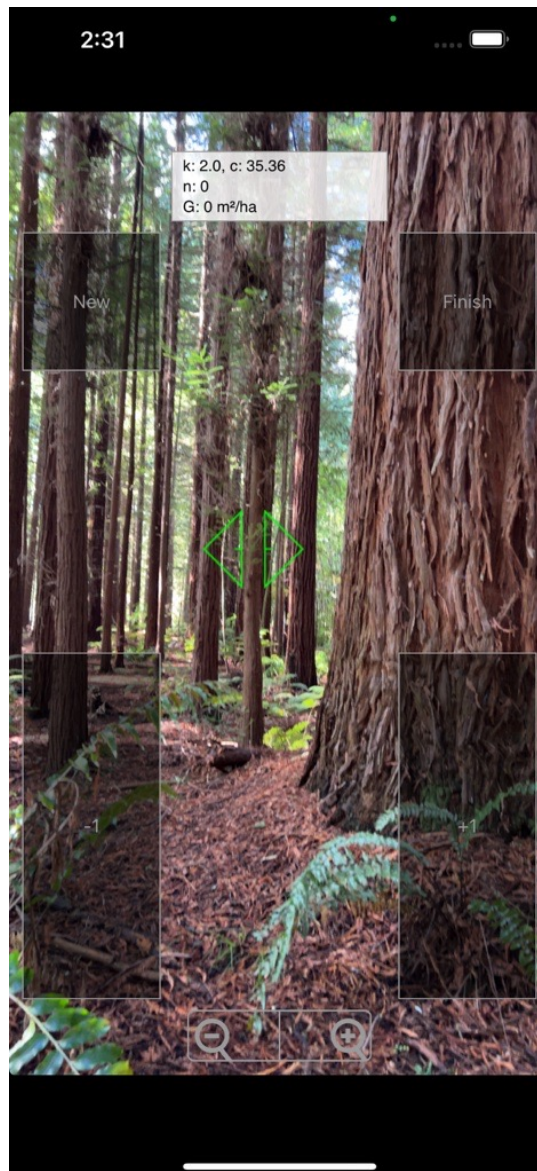


Figure 17. Screenshot of the MOTI app displaying the AR interface with the gauge and counters facilitating manual forest inventory.

1.8. Trees

- **Website:** <https://play.google.com/store/apps/details?id=com.forest.trees>
- **Platform:** Android
- **Recommended devices:** Android 7.0 and up
- **Owner:** Forest Monitoring Tools
- **Popularity:** 50,000+ downloads on the Google Play Store (Android)
- **Technologies used by phone to perform measurements:** AR/Camera, IMU
- **Country:** Japan

- **Year:** 2019
- **Last updated:** 2024
- **Pricing:** Free
- **Cloud storage:** No
- **Works offline:** Yes
- **Measurements/predictions:** Tree height
- **Ease of use and documentation:** Easy to use but requires additional tools to measure the distance to the tree, and to calibrate the app if the diameter should be measured too.
- **Comments:**

The app requires the user to enter the distance to the tree, then the tree height is computed through AR by marking the bottom, then the top of the tree (figure 18). The diameter can also be measured, but requires calibration.

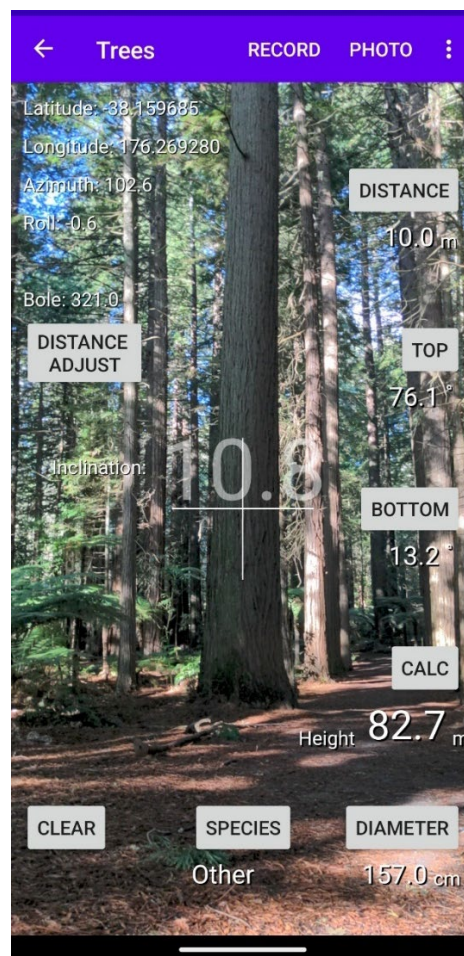


Figure 18. A screenshot of the Trees app displaying the AR interface after taking the measurements of one tree.

1.9. Trestima

- **Website:**

- <https://forestmeter.com/>
- https://play.google.com/store/apps/details?id=com.trestima.androidapp&hl=en_NZ
- <https://apps.apple.com/nz/app/trestima/id1106886045>

- **Platform:** Android, Apple

- **Recommended devices:** Requires iOS 14.0 or later (Apple), or Android 8.0 or later

- **Owner:** Trestima Oy

- **Popularity:** 5,000+ downloads on the Google Play Store (Android)

- **Technologies used by phone to perform measurements:** AR/Camera, IMU

- **Country:** Finland

- **Year:** 2014 Android, 2016 Apple

- **Last updated:** 2025 Apple, 2023 Android

- **Pricing:** from 4.59 euros / ha or 1.5 euros / sample

- **Cloud storage:** Yes

- **Works offline:** Photos can be taken, but the computation is done once the photos can be uploaded to Trestima's servers.

- **Measurements/predictions:** Basal area, number of trees per hectare, species, and volume, diameter and height ranges depending on the forest.

- **Ease of use and documentation:** The app is reasonably easy to use, but not very well documented.

- **Comments:**

The app works by having the user take sample photos of the forest (figure 19, 20), at least 10 recommended per stand. The app then detects stems and species from the photos and estimates the basal area and number of trees per hectare. Trestima has operators checking species identification when needed. The distribution of diameters and heights for each species can also be estimated, depending on previous data and models from similar forests being available to Trestima, or by using a measurement stick visible on the photos. The app can also track the user's path and photo coordinates. It is also possible to add different annotations, either in text or as drawing or polygons on the map. Full reports are available on Trestima's website once the photos have been uploaded and the computation done.

Sandim et al. ⁽¹⁾ found that the app fails to detect stems in some types of forest, and was the least accurate when compared to the Arboreal Forest and Katam apps (also described in this report). The authors noted however that it was also the least demanding app of the three to use, only requiring 10 photos to be taken for a given plot. Vastaranta et al. ⁽¹²⁾ reported that the app produces results with "reasonable accuracies". Other studies ^(11, 13, 14, 15, 16) found similar results.

Tests done by the authors with a Google Pixel 5 in a coast redwood forest found that the app was not very easy to navigate on a phone, and seems more designed to be used on a tablet. The app detected stems well, but mistakenly identified redwood stems as *Pinus radiata*, likely due to Trestima's New Zealand models not expecting redwood.

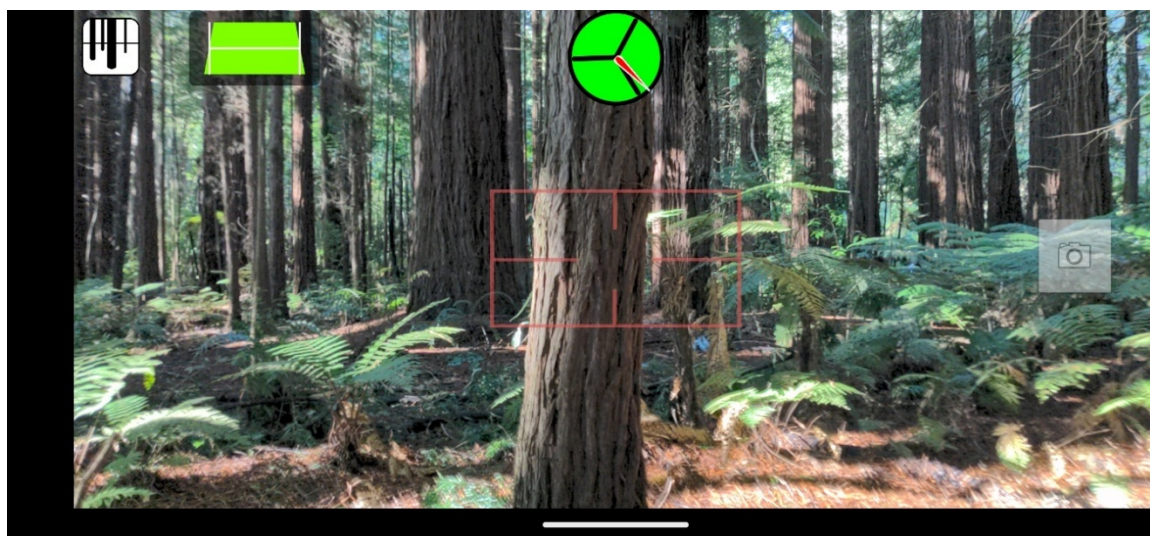


Figure 19. Screenshot of the Trestima app while taking photos used to compute measurements. The green indicators on the AR interface help using the right angles.



Figure 20. Screenshot of the Trestima app displaying the trajectory of the user and the positions of the photos that have been taken.

2. Applications that use phone or computer vision technologies for log pile measurement

2.1. (Arboreal) Tree Scanner

- **Website:** <https://arboreal.se/en/tree-scanner>
- **Platform:** Most likely Apple-exclusive due to lidar use
- **Owner:** Arboreal AB
- **Technologies used by phone to perform measurements:** AR/Camera, lidar, RFID
- **Country:** Sweden
- **Year:** Not released yet
- **Measurements/predictions:** Tree log measurements and tracking
- **Comments:**

This app is not released yet, but it is being developed by the same company as the first two apps described in section 1.

This app will use AR as well as the lidar sensor to measure logs. It will also integrate with external RFID readers to identify and track logs, and Hitman 220 sensors to measure wood quality on the spot.

2.2. iFOVEA Pro

- **Website:**
 - https://www.fovea.eu/ifovea_pro_woodpile_measurement?language=en
 - https://play.google.com/store/apps/details?id=eu.fovea.ifoveameasurement&hl=en_NZ
 - <https://apps.apple.com/nz/app/ifovea-pro/id859941143?platform=iphone>
- **Platform:** Android, Apple
- **Recommended devices:** Requires iOS 15.6 or later, or Android 7.0 or later
- **Owner:** palos GmbH
- **Popularity:** 10,000+ downloads on the Google Play Store (Android)
- **Technologies used by phone to perform measurements:** Camera
- **Country:** Germany

- **Year:** 2016 for Android, 2022 for Apple
- **Last updated:** 2024 for Android, 2025 for Apple
- **Pricing:** 150 Wood Units (cubic metres of timber): \$39.99 NZD, 1500 Wood Units: \$399.99 NZD, 2500 Wood Units: \$599.99 NZD
- **Cloud storage:** Possibility to synchronise the data with the company's online platform "Forest Management System"
- **Works offline:** Yes
- **Measurements/predictions:** Solid cubic meters, cubic meters (gross and net), number of logs.
- **Comments:**

From one or several photos of a log pile, the app predicts the number of logs, their diameters, and the pile's volume. The user has to enter the width of the front of the pile.

2.3. LogStackLIDAR

- **Website:**
 - <https://dalgas.com/en-gb/services/forests-trees/logsystems/logstacklidar>
 - <https://apps.apple.com/nz/app/logstacklidar/id1641890082>
- **Platform:** Apple
- **Recommended devices:** iPhone Pro and iPad Pro (2020 and newer) for lidar
- **Owner:** Dalgas/Hedeselskabet
- **Technologies used by phone to perform measurements:** AR/Camera, lidar
- **Country:** Denmark
- **Year:** 2022
- **Last updated:** 2025
- **Pricing:** Free demo, basic cost including 6,000 m3: 1,500.00 euros ; for more see <https://dalgas.com/en-gb/services/forests-trees/logsystems/prices>
- **Cloud storage:** Yes
- **Works offline:** Yes, but computation is performed in the cloud

- **Measurements/predictions:** Total stack volume, solid volume and total number of logs.

- **Comments:**

The app uses the lidar sensor of Apple pro devices to measure logs and log piles. The detected logs are displayed in real time through AR.

2.4. TimberID

- **Website:**

- https://www.deeplai.com/technology/mobile_apps/
- <https://apps.apple.com/nz/app/timberid/id1627463448>

- **Platform:** Apple
- **Recommended devices:** iPhone Pro and iPad Pro (2020 and newer) for lidar
- **Owner:** Deeplai
- **Technologies used by phone to perform measurements:** AR/Camera, lidar
- **Country:** Poland
- **Year:** 2023
- **Last updated:** 2025
- **Pricing:** Free for basic features, "advanced access" for \$39.99 NZD; more offers depending on client and needs.
- **Cloud storage:** Yes
- **Works offline:** Yes, but some of the computation is done in the cloud
- **Measurements/predictions:** Diameter and length of individual logs, and total pile volume.
- **Comments:**

The app uses the lidar of Apple devices to measure logs and log piles. The app offers more features, such as the decoding of markings on logs for tracking.

2.5. Timbeter

- **Website:**

- <https://timbeter.com/timbeter/>

- https://play.google.com/store/apps/details?id=ee.timberdiameter&hl=en_NZ
- <https://apps.apple.com/us/app/timbeter/id1249312902>
- **Platform:** Android, Apple
- **Recommended devices:** Requires iOS 13.0 or later, or Android 7.0 or later
- **Owner:** Timbeter Ltd
- **Popularity:** 50,000+ downloads on the Google Play Store (Android)
- **Technologies used by phone to perform measurements:** Camera
- **Country:** Estonia
- **Year:** 2015 for Android, 2017 for Apple
- **Last updated:** 2025
- **Pricing:** 10 first measurements free. No official pricing available, but an article from 2018 reports 125 euros per month or 2 euros/photo (https://archive.pfbc-cbfp.org/news_en/items/Forestworld-magazine.html).
- **Cloud storage:** Yes
- **Works offline:** Yes
- **Measurements/predictions:** Number of logs, plus volume and diameter of each log.
- **Ease of use and documentation:** Well documented, but seems to require some training.
- **Comments:**

From a photo of a log pile, the app predicts the number of logs, as well as the volume and diameter of each log. Requires a measurement stick visible on the photo.

2.6. Trestima Stack

- **Website:**
 - <https://www.trestima.com/w/en/stack-measuring-system/>
- **Owner:** Trestima Oy
- **Technologies used by phone to perform measurements:** Camera
- **Country:** Finland
- **Measurements/predictions:** Total volume of the pile, DBH distribution of logs, total log count, automatic coefficient factor for density, pile dimensions, location of the pile on map.

- **Ease of use and documentation:** Very little information found.
- **Comments:**

Trestima presents this app on their website, but no information is available on the availability or pricing. The app seems to compute all the estimations from a photo of a log pile. Requires a measurement stick visible on the photo.

3. Applications that facilitate manual entry for forest inventory

3.1. Arbornote

- **Website:**
 - <https://www.arbornote.com/>
 - https://play.google.com/store/apps/details?id=com.cognetiveSystems.ArborNote&hl=en_NZ
 - <https://apps.apple.com/nz/app/arbornote-for-tree-care-pros/id1065664362?platform=iphone>
- **Platform:** Android, Apple
- **Owner:** Cognetive Systems, Inc.
- **Country:** US
- **Year:** around 2013
- **Last updated:** 2025
- **Type:** Manual entry, targeted towards arborists and tracking of urban trees. GPS and image tagging of trees.

3.2. Arborpro

- **Website:**
 - <https://arborprousa.com/arborpro-software/>
 - <https://apps.apple.com/nz/app/arborpro/id1561907328>
- **Platform:** Apple
- **Owner:** ArborPro, Inc.

- **Country:** US
- **Year:** 2023
- **Last updated:** 2025
- **Type:** Manual entry, targeted towards arborists and tracking of urban trees.

3.3. FirsDC

- **Website:**
 - <https://firs.com.au/Home/FirsDcLearnMore>
 - <https://play.google.com/store/apps/details?id=au.com.firs.firsdca&hl=en>
- **Platform:** Android
- **Owner:** Universal Forest Systems Pty Ltd
- **Country:** Australia
- **Year:** 2018
- **Last updated:** 2024
- **Type:** Manual entry for forest inventory data.

3.4. Forest Metrix PRO

- **Website:**
 - <https://forestmetrix.com/>
 - <https://forestmetrix.com/downloadpro/>
- **Platform:** Apple
- **Owner:** Forest Metrix
- **Country:** US
- **Year:** 2014
- **Last updated:** Unknown
- **Type:** Manual entry, targeted toward arborists and tracking of urban trees.

3.5. Forestree

- **Website:** <https://forestree.app/>
- **Platform:** Android and Apple
- **Owner:** Forestree Pty Ltd
- **Country:** Australia
- **Year and last updated:** Unknown
- **Type:** Manual entry, targeted towards arborists and tracking of urban trees. GPS and image tagging of trees.

3.6. FScruiser

- **Website:**
 - <https://github.com/FMSC-Measurements/NatCruise/wiki/FScruiser-V3-User-Guide>
 - https://play.google.com/store/apps/details?id=com.FMSC.FScruiser&hl=en_NZ
- **Platform:** Android
- **Owner:** US Forest Service
- **Country:** US
- **Year:** 2022 (version V3)
- **Last updated:** 2024
- **Type:** Cruiser, manual entry.

3.7. MSUES TreeMetrics

- **Website:**
 - <https://techoutreach.extension.msstate.edu/apps/msues-treemetrics>
 - <https://apps.apple.com/us/app/msues-treemetrics/id963732737>
- **Platform:** Apple
- **Owner:** Mississippi State University Extension Service
- **Country:** US
- **Year:** 2015

- **Last updated:** 2015
- **Type:** Manual entry, targeted towards arborists and tracking of urban trees. GPS and image tagging of trees.

3.8. SilvAssist Mobile

- **Website:**
 - <https://f4techconsulting.blob.core.windows.net/supportmanuals/SilvAssistMobile%20Users%20Manual.pdf>
 - https://play.google.com/store/apps/details?id=com.f4tech.silvassistmobile&hl=en_NZ
 - <https://apps.apple.com/nz/app/silvassist-mobile/id1573671076>
- **Platform:** Android, Apple
- **Owner:** ForesTech Consulting, Inc.
- **Country:** US
- **Year:** 2021
- **Last updated:** 2024
- **Type:** Built on ArcGIS, cruiser

3.9. Tree Risk Assessment - Level 1

- **Website:**
 - <https://texasforestinfo.tamu.edu/MobileApps/TreeRisk/>
 - https://play.google.com/store/apps/details?id=com.tfs.level1.android&hl=en_NZ
 - <https://apps.apple.com/us/app/tree-risk-assessment-level-1/id917769134>
- **Platform:** Android, Apple
- **Owner:** Texas A&M Forest Service (TFS)
- **Country:** US
- **Year:** 2014
- **Last updated:** 2022
- **Type:** Manual entry for pre and post storm risk inventory.

3.10. Tree Count (- Tree Inventory)

- **Website:**
 - <https://texasforestinfo.tamu.edu/MobileApps/TreesCount/>
 - <https://apps.apple.com/us/app/trees-count-tree-inventory/id1281104625>
 - https://play.google.com/store/apps/details?id=edu.tfs.tamu.tfs_tree&hl=en
- **Platform:** Android, Apple
- **Owner:** Texas A&M Forest Service (TFS)
- **Country:** US
- **Year:** 2017
- **Last updated:** 2022 Apple, 2025 Android
- **Type:** Manual entry, targeted towards arborists and urban forest management.

4. Conclusion

This review found nine apps that use phone or computer vision technologies for tree or forest measurements, six apps that use those technologies for log and log pile measurements, and ten apps designed to facilitate manual entry of forest inventory.

In the last decade, sensors in smartphones have become more common and varied, as well as more accurate. This has enabled app developers to leverage those sensors to create apps that perform a range of tree and forest measurements, that don't require the users to use any other equipment or perform any manual measurements.

The apps presented in the first part of this report often require forest conditions comprising sparse understory and straight stems of average size to work well. Previous reviews, however, found that most of them can compute results of an accuracy comparable to that of manual measurements. Most of the apps are easy to use, but require some training and testing to get quick and accurate measurements.

Overall, Arboreal Forest appears to be the most complete, user-friendly, industry-friendly, and accurate app among the apps using phone or computer vision technologies for tree and forest measurements. The Arboreal AB company that developed the app also appears to constantly improve its apps based on client feedback and requests.

In summary, the implications for industry are that these apps are becoming increasingly accurate, easy to use and functional as the technology develops, enabling a growing number of measurements for a range of applications. The groups

developing those apps are often small and responsive, with a lot of potential for customisation of these apps to suit the requirements of NZ foresters.

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